PX4128 Data Analysis Block A

Background Reading

Written by Dr Paul Clark
adapted by Prof Haley Gomez

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Contents

Chapter 1

Overview of Data Analysis and Probability Theory

1.1 What do we mean by 'Data Analysis'?

Although Data Analysis is very broad term, it typically results in us trying to do one of the following 3 tasks.

- **Estimation of parameter values** Which of the possible values for a given parameter is best represented by the data? This type of analysis typically occurs when the general form of the model is clear, but we need to quantify the underlying details. One could arguably propose that much of modern cosmology is in this category – don't tell the cosmologists that I said that...
- **Prediction of data values** Given what we know about a model, can we use it to predict unseen data – i.e. parts of the parameter space that are yet to be observed? Although the obvious examples here include predicting the weather, or the flight of a cricket ball, note that the applications need not be temporal! For example, we could ask, how well will a temperature sensor record temperature, when exposed to an unobserved (untested) environment?
- **Model comparison** Given some data, can we determine which, of any (!), or the competing models/theories for a particular system are doing the best job of describing reality? Perhaps more importantly, can the data actually discriminate (does it have the power) or is a different experimental approach needed?

1.2 Probably useful: probability overview

- We are going to be using the concept of probability a lot in this course, so it is best to first review the basic ideas behind probability theory, and get used to the notation.

Unfortunately, there are many notations; this course will adopt one, but we will also mention the others, so that you can recognise them when reading further.

- An **experiment** results in a set of outcomes, which we will call Ω. This can be a discrete set of outcomes, such is in the classic coin toss, $\Omega = \{H, T\}$, or the roll of a die, $\Omega = \{1, 2, 3, 4, 5, 6\}$. However in many real life experiments, Ω, which is referred to as the **outcome space**, or **event space**, can have an infinite continuum of values. We will return to this idea later, but for the moment we will consider the discrete events to help outline the basic properties of probability.

- Returning to the coin toss, we can say that a 'fair' coin, will have a probability of heads of $P(H) = 0.5$, and a probability of tails of $P(T) = 0.5$. Each outcome of the experiment of tossing the coin, $\Omega = \{H, T\}$, are thus equally likely. Similarly, for a die roll, $\Omega = \{1, 2, 3, 4, 5, 6\}$, with $p(i) = 1/6$. When an experiment has m equally likely outcomes, the probability of any outcome x is then,
$$P(x) = \frac{\#x}{m}, \tag{1.1}$$
where $\#x$ is the number of times that x occurs. For example, consider the more complicated case where we toss 3 different coins together, a 50p, a 20p and a pound coin. The outcome space is,
$$\Omega = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}. \tag{1.2}$$
Assuming all outcomes were equally likely, then the probability of any one is $1/8$.

- Not all outcomes are equally likely. For example, we could ask what the probability of the event that our 3-coin toss comes up with n heads, so the outcome space is then $\Omega = \{0, 1, 2, 3\}$. The outcome $\omega \in \Omega$ in this new experiment, is just given by considering the outcomes in our previous 3-coin experiment, but ignoring which exact coin lands H/T:

  1. $\omega = 1$: n = 0, corresponds to TTT
  2. $\omega = 2$: n = 1 corresponds to HTT, THT, or TTH
  3. $\omega = 3$: n = 2 corresponds to HHT, HTH, or THH
  4. $\omega = 4$: n = 3 corresponds to HHH

  Thus $P(n = 0) = P(n = 3) = 1/8$ while $P(n = 1) = P(n = 2) = 3/8$.

- The above uses of P hide an important aspect of probabilities: $P(x_i)$ is **normalised**, such that the sum of the probabilities of all possible events adds up to 1,
$$P(X) = \sum_i^N P(x_i) = 1, \tag{1.3}$$
where $X = (x_i, \ldots, x_N)$. Technically this in only true for either finite or countably infinite outcome spaces. If the outcome space is truly uncountably infinite, then the definition of probability has to be relaxed.

- At this point we should introduce the 'axioms' of probability. They are

  Axiom 1 $0 \leq P(A) \leq 1$

  Axiom 2 $P(\Omega) = 1$

  Axiom 3 For mutually exclusive (pairwise disjoint) events, $A_1, A_2, \ldots,$

$$P(A_1 \cup A_2 \cup A_3 \ldots) = P(A_1) + P(A_2) + P(A_3) + \ldots. \tag{1.4}$$

  Here, the symbol $\cup$ denotes 'or', i.e. the probability of outcome A_1, A_2, A_3, etc.

- The axioms permit us to work out the probability of event A **not** occurring,

$$P(A^c) = 1 - P(A). \tag{1.5}$$

  This result can be generalised. For example, for any two events C, D,

$$P(C \cup D) = P(C) + P(D) - P(C \cap D), \tag{1.6}$$

  where now the symbol $\cap$ denotes 'and' (also written simply $P(CD)$ or $P(C, D)$ in the literature). A simple way to think of this is that the probability of getting either C or D is just the sum of the chances of getting either, $P(C) + P(D)$, minus that chances of getting both at the same time, $P(C \cap D)$. The last part is important since we're asking for the probability of **either** C **or** D, not both! The best way to see this is by considering the diagram in Figure 1.1.

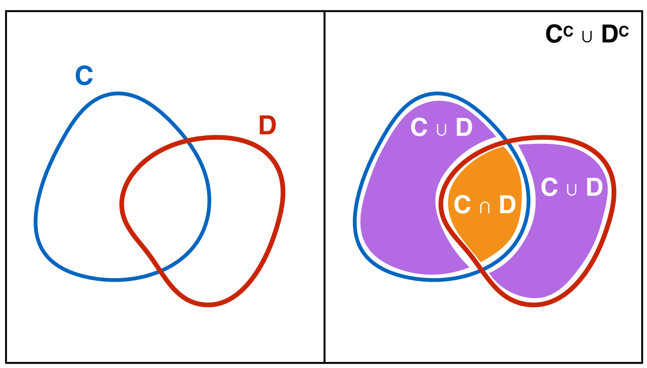

Figure 1.1:

- Note that the above expression is valid, whether or not the events are mutually exclusive (disjoint). That is the purpose of the 'and' term, since it subtracts the probability that both the events occur at the same time. How to calculate $P(C \cap D)$? Well, if the events are independent (i.e. do not depend on the other event occurring), then

$$P(C \cap D) = P(C)P(D), \tag{1.7}$$

  i.e. just the product of the probability of the two events.

- We now have enough knowledge of the probability basics to consider **conditional probabilities**. Technically speaking, **all** probabilities are conditional. For example, the probability that my coin will land either heads or tails is conditioned by the probability that the coin will land. Or indeed have a heads and a tails! In a less contrived example, the probability that an astronomer is observing a specific type of star, say an A2 star, is first conditioned on the probability that what they are observing is indeed a star, and not another astronomical phenomenon. So conditional probabilities are important. They are denoted in the following way: the probability of event A given condition (or event) B is $P(A|B)$.

- Take a classic example: what is the probability of randomly drawing the Queen of Spades (QoS) from a well-shuffled, true pack of cards? Well, there are 52 cards in total, so the probability of drawing **any** card (C) is simply $P(C) = 1/52$. The probability of drawing our desired QoS is then $P(QoS) = 1/52$. Now consider that the dealer is truthful, and tells you that the card you have just drawn is a face (F) card. Now what is the probability $P(QoS)$? Well, first, the probability of drawing a face card that's also the QoS is given by $P(QoS \cap F)$. But since we know that the card is a face, our probabilities for $P(QoS)$ and $P(F)$ are wrong in the sense that they were derived by diving by all cards – 1/52, or 12/52, since there are 12 face cards in the pack. Instead we want to renormalise our probabilities to the region of outcome space where F is true, so we need to divide by $P(F)$. More generally, for two independent events A and B, we have,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \tag{1.8}$$

A simple way to visualise this result is to consider the areas in Figure 1.2.

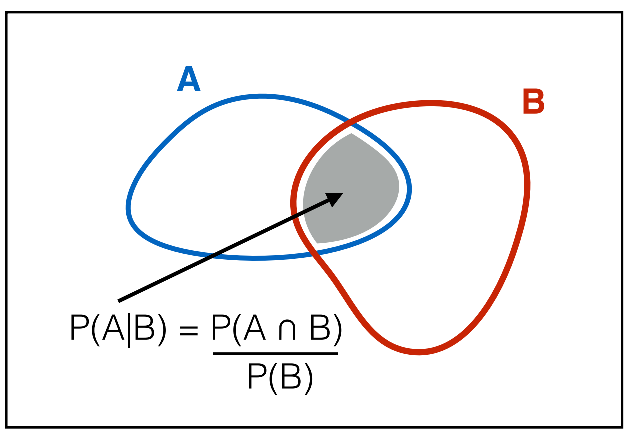

Figure 1.2:

1.3 Bayes Theorem

- The famous Bayes Theorem is derived from considering the conditional probability in 1.8 above. Given that $P(A \cap B) = P(B \cap A)$, then we can use 1.8, to write,

$$P(A|B)P(B) = P(B|A)P(A) \tag{1.9}$$

such that,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \tag{1.10}$$

which the standard form of Bayes Theorem. One can generalise the denominator in 1.10 by considering that,

$$P(B) = P(B \cap A) + P(B \cap A^c) = P(B|A)P(A) + P(B|A^c)P(A^c) \tag{1.11}$$

to give the result,

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)}, \tag{1.12}$$

- Bayes Rule is extremely powerful, and allows us to deal with complex problems quite simply.

- Often Bayes Rule is being used to determine the probability of some model parameter θ (or set of model parameters) given data D. The standard way to write Bayes Rule is then,

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)} \tag{1.13}$$

where, the terms have the following names and meanings:

$P(\theta|D)$ the 'posterior' - the probability of model parameter θ being true, given the data

$P(D|\theta)$ the 'likelihood' - given model parameter θ what is likelihood of obtaining the data

$P(\theta)$ the 'prior' - the probability of the model parameter θ being 'true'

$P(D)$ the 'evidence' - the probability of getting the data, give all possible model parameter values (θ and others!)

- People feel uncomfortable about priors, since the end result (the posterior) depends on a 'guess' by the person doing the analysis, and feel that Bayesian reasoning is not scientific or exact enough. Indeed, different modellers may also have different opinions about what the prior should be, thus presenting different posteriors in their papers. However, it is important to realise that even the standard 'frequentist' view of the world, which traditionally uses only the likelihood to determine the posterior is also adopting a prior: they are implicitly assuming that the prior is flat! This means that they assume that all values of θ are equally likely. Clearly this is not only wrong, but extremely limiting. For example, if you see someone across the road in your home town that looks like your friend, then the chances are it is your friend. If you see someone that looks like your friend when you're visiting a remote island that is far away from your home, the chances are it is not your friend. The prior allows you to account for this.

- However it should also be accepted that the prior is, often, subjective, and people will argue about what it should be. Therein lies a strength – researchers must declare what their prior is and test how sensitive their results are to the prior. But in other cases, the prior will be informed by previous experimental knowledge, for which a careful study of the data has helped to constrain the values of $P(\theta)$.

- In light of newly arriving data, we can reapply Bayes Rule to get a new posterior. But what to use for the prior? Well, the posterior from the previous analysis! Hence,

  "Yesterdays posterior is tomorrow's prior"

  This 'feedback' loop makes Bayes Theorem particularly useful in machine learning, in which decision making needs to adapt to new information as it comes in.

So why is Bayesian analysis 'difficult'? In general, the 'evidence' – the integral in the denominator of Bayes Rule – can be difficult to evaluate. For analytic solutions, life can be made easier with a suitable choice of prior. For example, an **orthogonal prior** is one that has the same functional form as the likelihood. So provided you are free to alter the shape of the prior, or an approximation is sufficient, then choosing a prior with the same shape as the likelihood, can simplify the mathematics we will show an example of this later in the course. However, often the integral of the product of the likelihood and prior are not analytically tractable. In these circumstances, we need to resort to numerical methods for computing the integral. If the parameter space is not too large, then one can often simply grid the dependent variables and compute a rough integral from the sum of the functions at the variable points. However, when the parameter space becomes large (and this can occur surprisingly quickly!), then we need to resort to a different numerical method called a **Monte Carlo Markov Chain**. We will look at this method later in the course.

1.4 So what is probability?

- In the above examples of the coin flipping, die rolling and card selecting, we introduced the notion that the probability of a particular outcome or event is simply the number of times that event occurs, divided by the number of all possible outcomes.

- But say you have a coin, and you want to know $P(H)$ – how do you proceed? You could guess that the coin is 'fair' and assign 0.5 to outcome heads/tails. This is essentially what we did above. But is the coin fair? One way you could test this is to perform lots of experiments (coin flips) and keep track of the outcome. If you do enough of these, eventually you will get an empirical measure,

$$P(H) = \frac{n_H}{n_{flips}} \tag{1.14}$$

where n_H is the number of heads that appeared in the experiment and n_{flips} is the number of times you flipped the coin (and counted the result). But when do you stop? Well, that depends on how accurately you want to know $P(H)$, and we will look into this later. But for now, we will simply note that this type of determination of P is **frequentist**, in that the probability is defined by counting the instances of occurrence.

- However, what about the probability that it will rain tomorrow? You can see straight away that such a probability is more difficult to define. In fact, the use of Bayes Theorem, and in particular the prior, introduces a much more vague idea that P represents the belief that something will occur.

Chapter 2

Some standard PDFs, the mean, variance, and addition of errors

This chapter is a summary of Chapters 4 & 5 in Taylor (and closely follows them). For more details and examples, see the original text.

2.1 Probability density functions (PDFs)

- Up until now, we have only considered the probability of discrete events, such as a coin-flip resulting in heads, or a die turning up with a 1. However, probabilities can also be determined for continuous variables, for example, the probability that a child we be a certain height at a given age, or that the intensity in a spectrum will be a given value, or that a molecule will have a given velocity. In this case, the height $h(age)$ or intensity $I(\lambda)$, or velocity v, are all continuous variables.

- Just as the discrete probabilities for all possible events need to add up to unity, so too does the continuous probability. If a is a discretised version of b, then we can say,

$$\sum_i^N P(a_i) = 1 = \int_{b_{min}}^{b_{max}} p(b)\mathrm{d}b, \tag{2.1}$$

  where $p(b)$ is the **probability density function (PDF)** and is normalised to 1. From this, we can see that the probability of b lying in some interval $b_1 \rightarrow b_2$ is then given by,

$$p(b_1 \rightarrow b_2) = \int_{b_1}^{b_2} p(b)\mathrm{d}b \tag{2.2}$$

- It is important to note that in the case of a continuous variable, the PDF has **units that are the inverse of the dependent variables**, e.g. $p(b)$ has units of $1/b$, or, in the case of our example of the velocities of the molecules, $p(v)$ has units of s m^{-1} (i.e. inverse velocity).

- It is also possible for the PDF to be a function of more than 1 variable! For example there may exist a function p(x, y) that yields the probability of both x and y. We will see plenty examples of these later in the course, and note that this type of function may be a combination of discrete and continuous variables. Once again, the units are $1/xy$.

- But suppose we wanted to know just $p(x)$. How do we calculate this, given that we only have $p(x, y)$? If you consider the units of the problem, the answer becomes clear: we integrate or sum over y, i.e.

$$p(x) = \int p(x, y)\mathrm{d}y \tag{2.3}$$

  if y is a continuous variable, or

$$p(x) = \sum_y p(x, y) \tag{2.4}$$

  if y is discrete. This process is called **marginalising over y or integrating out y**. We could do the same to get $p(y)$. Marginalisation is extremely important, since it allows us to deal with **nuisance** parameters, that is, those that we don't know very well (or that are not well constrained).

- However marginalisation can also play another role. Consider that we want to know what the probability of a given value of y is, for a given value (or even range) of x. In probability notation, this is simply

$$p(y|x) = \frac{p(y, x)}{p(x)} = \frac{p(y, x)}{\int p(x, y)\mathrm{d}y}, \tag{2.5}$$

  in the case that y was a continuous variable. Here we see that marginalisation is a common feature in conditional probabilities. In fact, it is a common process for determining the denominator in Bayes Rule.

- As an example, consider that we have a function p(h, a) that describes the height h of people as a function of their age a. Let's say that this function was obtained from a study that fitted a curve to the histogram of heights of people in a given age group, and the ages were grouped by year (i.e. a histogram of all heights in each age 1, 2, 3, . . ., etc). In this case, the function is continuous in h, but discrete in a. Now say that we want to know the probability of finding someone of age 23 years old, with a height in the range $h_1 \to h_2$. We can now evaluate this from,

$$p(h_1 \to h_2|a_{23}) = \frac{p(a_{23}, h_1 \to h_2)}{p(a_{23})} = \frac{\int_{h_1}^{h_2} p(a_{23}, h)\mathrm{d}h}{\int_{h_{min}}^{h_{max}} p(a_{23}, h)\mathrm{d}h}. \tag{2.6}$$

  In the denominator, we take into account that the people of 23 can have a wide range of heights between h_{min} and h_{max}.

2.2 Data descriptors: the mean and variance

In Section 2.1, we discussed the idea of PDFs and how they are used. In this section, we will look into the idea of PDFs a little further, and take a look at the mean and variance.

- Imagine that we are trying to measure the length x of a snake. Unfortunately, this is pretty tricky, as the snake keeps moving around, and so our individual results are subject to an error that is difficult to estimate *a priori*. In the end, we decide to take 10 measurements and recover the following values:

  $$26, 24, 26, 28, 23, 24, 25, 24, 26, 25$$

  Typically, the **best guess** for the length of the snake would just be the **mean** defined as,

  $$\hat{x} = x_1 + x_2 + \cdots + x_N = \frac{\sum x_i}{N}, \tag{2.7}$$

  where we have used $\sum x_i$ as a shorthand for $\sum_i^N x_i$.

- Now that we have the mean as the best guess, what about an estimate of the **error** or **uncertainty** in each individual value? One natural method is ask what the difference is between the best value, $\hat{x}$, and the individual points, $d_i = x_i - \hat{x}$. In this case, we would get 10 values for the difference – but we want a single number! What about taking the mean, such that

  $$\hat{d} = \frac{\sum x_i - \hat{x}}{N} \tag{2.8}$$

  The problem here is that d_i can be positive and negative (points lie either side of the mean), so the sum could, in certain cases, reduce to zero. Better, is to consider the squares of the difference, so that the terms don't cancel. This is more commonly referred to as the **standard deviation**, and denoted by the symbol σ_x. It is given by,

  $$\sigma_x = \sqrt{\frac{1}{N}\sum_i^N (x_i - \hat{x})^2}. \tag{2.9}$$

  Strictly speaking, the factor $1/N$ in 2.9 should be replaced by $1/(N-1)$. This has to do with the idea of **degress of freedom** in formal statistics (which we will not derive here), and comes from the fact that σ_x is a function of $\hat{x}$, which was already calculated from the data. Since σ_x is then the second parameter to be calculated from the data, we require a $1/(N-1)$, giving,

  $$\sigma_x = \sqrt{\frac{1}{N-1}\sum_i^N (x_i - \hat{x})^2}. \tag{2.10}$$

  Note that the square of the standard deviation is termed the **variance**,

  $$\sigma_x^2 = \frac{1}{N-1}\sum_i^N (x_i - \hat{x})^2. \tag{2.11}$$

- So we now know what the error is in an individual measurement x, but what about the error of the best guess, $\hat{x}$? This is given by the so-called **standard deviation of the mean**, defined as,

$$\sigma_{\hat{x}} = \sigma_x/\sqrt{N}. \tag{2.12}$$

The equation makes sense: since our best guess $\hat{x}$ is a combination of the individual estimates x_i, we expect the error in $\hat{x}$ to be less than these individual measurements. The factor $\sqrt{N}$ is important, since it tells us that although we can make $\hat{x}$ better by taking more measurements, the result get better slowly: a factor 10 improvement in our error estimate requires 100 more measurements!

2.3 Histograms as a stepping stone to PDFs

- Histograms are often a useful way to represent data, and in many cases, the first step to creating an empirical PDF from the results of an experiment. As an example, we will consider our set of 10 measurements of the snake above. To make it clear what we are doing, we will first order the data

23, 24, 24, 24, 25, 25, 26, 26, 26, 28.

We can then quickly see that we could represent this data in the following way (we don't need to sort the data first, but it makes it easier to see here!) : We can then

Table 2.1: Results from the snake measuring experiment

Measured length of snake x_k	23	24	25	26	27	28
Instances of length x_k	1	3	2	3	0	1

rewrite the mean in way that makes things a little more convenient in the end:

$$\hat{x} = \frac{\sum_k x_k n_k}{N} \tag{2.13}$$

where n_k is the number of instances of the **different** measurement x_k, as given in the bottom row of table 2.1. Expression 2.13 is a **weighted sum**, since each value is weighted by the number of times it occurs. Note also that

$$\sum_k n_k = N \tag{2.14}$$

- Another way of thinking about this is that each result x_k occurs a certain fraction of times – e.g. 24 is obtained in 3/10 of all measurements, while 28 is obtained only in 1/10, and so we can introduce the fraction,

$$F_k = \frac{n_k}{N}. \tag{2.15}$$

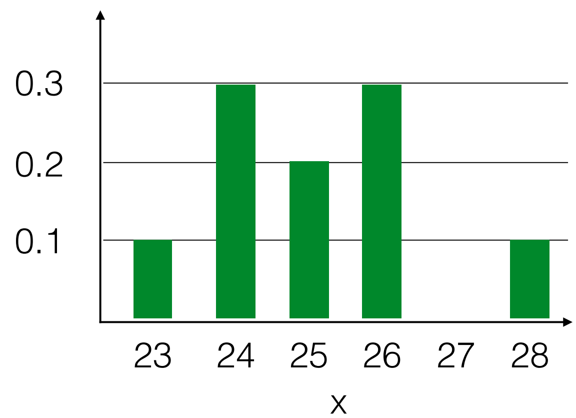

Figure 2.1: Histogram of the analysis of a snake measuring experiment showing F_k as a function of snake length x

We can then rewrite the mean as:

$$\hat{x} = \sum_k x_k F_k, \tag{2.16}$$

and once again note that $\sum_k F_k = 1$.

- By creating a bar graph of F_k against x_k we obtain a **histogram** of the data from the experiment, which can be a useful way of quickly seeing the underlying shape of the distribution of results. An example of this for the snake data is shown in 2.3. However, it is also possible that we can have data that are not exactly 23 or 24, but say 23.6 and 24.3. How do we accommodate these? The answer is to have bins that have a finite width, rather than the infinitely thin bars represented by F_k. For example, suppose our snake measuring experiment resulted in the following data:

  26.4, 23.9, 25.1, 24.6, 22.7, 23.8, 25.1, 23.9, 25.3, 25.4

  We could then represent this data in the following bins: Rather than representing a bin

Table 2.2: More accurate results from the snake measuring experiment

Bin (length range)	22 to 23	23 to 24	24 to 25	25 to 26	26 to 27	27 to 28
Observations in bin	1	3	1	4	1	0

by its height F_k, we now need to also define some width, Δ_k. We therefore define the **area** of the bin to represent the fraction of measurements that fall within the bin,

$$f_k \Delta_k = \text{fraction of measurements in the } k\text{th bin} \tag{2.17}$$

An example of this type of histogram is shown in 2.3.

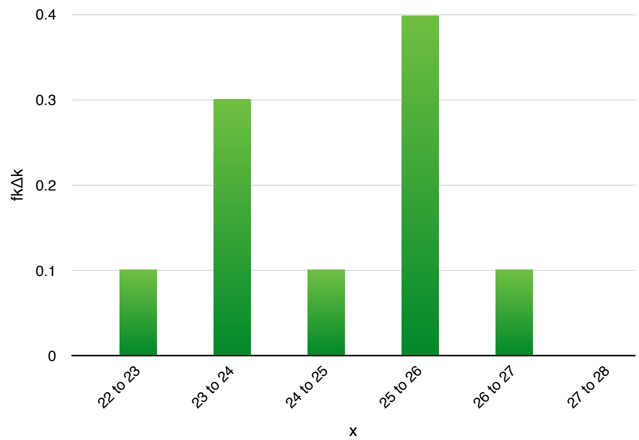

Figure 2.2: Histogram of the analysis of a snake measuring experiment with better data.

- As the number of measurements approaches infinity, two things occur. The first, is that that distribution given by our histogram is said to approach the **limiting distribution**. The second is obviously that the component $f_k \to f(x)$ and $\Delta_k \to dx$. Just as the sum of $f_k\Delta_k$ is unity for all k,

$$\int_{-\infty}^{\infty} f(x)dx = 1. \tag{2.18}$$

Thus, in the limit of many observations, the histogram approaches the PDF of the limiting distribution; histograms can be thought of as a crude PDF.

- We can now introduce a more formal definition of the mean:

$$\hat{x} = \sum_k x_k\, F_k = \sum_k x_k\, f_k\, \Delta_k \overset{N\to\infty}{=} \int_{-\infty}^{\infty} x\, f(x)\, dx \tag{2.19}$$

and similarly for the variance,

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x-\hat{x})^2\, f(x)dx. \tag{2.20}$$

Note here that the problems we had with $1/N$ & $1/N-1$ are irrelevant since we are now working in the regime where $N \to \infty$.

- Another important property of any PDF is its **cumulant** distribution, which is given, as a function of the dependent variable x as,

$$F(x') = \int_{-\infty}^{x'} f(x)dx. \tag{2.21}$$

Once again, the lower bound need not be $-\infty$, but could simply be the lower limit over which the limiting distribution is valid. The cumulant tells us the **percentile** that x' represents. For example, if $F(x') = 0.6$, then 0.6 (or 60%) of the area under $f(x)$ would lie in the range $\leq x'$.

- The **median** denotes a special case of x' for which $F(x) = 0.5$ – the '50th percentile'. This value represents the midpoint of the data, since there are as many data values below x' as above.

2.4 The Normal Distribution

- The **central limit theorem** states that if a quantity is subject to many small, but independent, random processes, the spread of the quantity can be described by a bell-like curve, known as a **Gauss function** or **normal distribution**, which has the form,

$$N(x) \propto e^{-x^2/2\sigma^2} \tag{2.22}$$

which is centred on $x = 0$, and has a width controlled by σ. To centre the function around some other value, the numerator in the exponential is replaced with $x - x_0$, where x_0 now defines the centre.

- If the measurement of a particular quantity is subject to many, independent and random errors, then the central limit theorem also allows us to use the normal distribution to model the quantity's errors. Although it is not alway the case that the errors are normal, it is often a good enough approximation, and one which makes the mathematics of the error analysis significantly simpler – as we shall now explore in the rest of this section.

- If we are to use 2.22 as a PDF of a quantity or its associated errors, we first have to normalise the function, such that it satisfies,

$$\int_{-\infty}^{\infty} N(x)dx = 1. \tag{2.23}$$

We first remove the proportionality in 2.22 and introduce a constant C, such that,

$$N(x) = C\, e^{-(x-x_0)^2/2\sigma^2}. \tag{2.24}$$

Note that C only serves to the move the curve up an down in y, but leaves the shape and centring undisturbed; it obviously changes the area under the curve though, which is the whole point in the normalisation. To evaluate the integral, we make a change of variable, by setting $(x - x_0)/\sigma = z$, such that $dx = \sigma dz$ to get,

$$\int_{-\infty}^{\infty} N(z)dz = C\sigma \int_{-\infty}^{\infty} e^{-z^2/2}dz. \tag{2.25}$$

This is a standard in physics, and has the result,

$$\int_{-\infty}^{\infty} e^{-z^2/2}dz = \sqrt{2\pi}, \tag{2.26}$$

which yields the value for the normalisation $C = 1/\sigma\sqrt{2\pi}$. We can then write the final form for the normal distribution as,

$$N_{x_0,\sigma}(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-x_0)^2/2\sigma^2}. \tag{2.27}$$

Measurements whose **limiting function** is given by 2.27 are said to be **normally distributed**.

- So what is the mean of the normal distribution? Following our definition of the mean of a PDF, given by 2.20 above, we need to evaluate,

$$\hat{x} = \int_{-\infty}^{\infty} x\, N_{x_0,\sigma}(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} x\, e^{-(x-x_0)^2/2\sigma^2} dx. \tag{2.28}$$

Again, this can be evaluated with a change of variables, replacing $x - x_0 = y$, such that $dx = dy$ and $x = y + x_0$. This results in,

$$\hat{x} = \frac{1}{\sigma\sqrt{2\pi}} \left(\int_{-\infty}^{\infty} y\, e^{-y^2/2\sigma^2} dy \ + \ x_0 \int_{-\infty}^{\infty} e^{-y^2/2\sigma^2} dy \right). \tag{2.29}$$

The first integral is zero, since although the exponential term is symmetric about $y = 0$, the y pre-factor is not, and so the points from $-y$ are exactly cancelled by those from $+y$. The second integral is the same as that we seen above, and is just $\sigma\sqrt{2\pi}$, which cancels with the term at the front, leaving us with,

$$\hat{x} = x_0, \tag{2.30}$$

which is what one would expect, given the shape of the curve!

- Similarly, we can compute the standard deviation by,

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x - \hat{x})^2 N_{x_0,\sigma}(x) dx \tag{2.31}$$

by noting that $\hat{x} = x_0$, and then by making the substitutions $x - x_0 = y$, and $y/\sigma = z$, and then integrating by parts. This gives,

$$\sigma_x^2 = \sigma^2. \tag{2.32}$$

So for the normal distribution, the standard deviation is given by the width parameter σ. The fact that that $\hat{x}$ and σ_x relate to the parameters that describe the normal distribution is one of the reasons why it is so commonly applied in data analysis.

- Since $N_{x_0,\sigma}(x)$ is a PDF, the probability of x lying in the range a to b is then given by,

$$\int_a^b N_{x_0,\sigma}(x) dx. \tag{2.33}$$

So what about the probability of lying within $\pm t\sigma$, where t is some real (positive) number? This is given by,

$$P(\text{within } t\sigma) = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_0 - t\sigma}^{x_0 + t\sigma} e^{-(x-x_0)^2/2\sigma^2} dx. \tag{2.34}$$

Once again, substitution of $(x - x_0)/\sigma = z$, with $dx = \sigma dz$ and now limits of $-t$ to t, we have,

$$P(\text{within } t\sigma) = \frac{1}{\sqrt{2\pi}} \int_{-t}^{+t} e^{-z^2/2} dz. \tag{2.35}$$

Although a common integral in physics, and is known as the **error function**. Unfortunately, it can not be evaluated analytically, however using a computer, it is possible to obtain the following values for the integral as a function of t,

Table 2.3: The probability that measurement x will fall within $t\sigma$ of the mean in the normal distribution.

t	0.25	0.5	0.75	1.0	1.5	2.0	2.5	3.0	3.5	4.0
$P(\text{within } t\sigma)$	0.2	0.38	0.55	0.68	0.87	0.954	0.988	0.997	0.9995	0.9999

- **Justification of the mean as the best estimate.** Imagine we took N measurements some value x, which we believe to follow a normal distribution with unknown parameters x_0 and σ. We want to find an estimate for the **true** values that describe this limiting function. We said before that this is simply the mean and standard deviation of our measurements, and given what we've seen above this makes sense. However let's prove it. First, consider that if $N_{x_0,\sigma}$ is the limiting function, then we can write the probability of obtaining our N measurements of x as,

$$P_{x_0,\sigma}(x_1, \ldots, x_N) = P(x_1) \times P(x_2) \cdots \times P(x_N) \tag{2.36}$$

$$P_{x_0,\sigma}(x_1, \ldots, x_N) \propto \frac{1}{\sigma^N} e^{-\sum(x_i - x_o)^2/2\sigma^2} \tag{2.37}$$

To find the best estimate for x_0 and σ, we are going to invoke the **principle of maximum likelihood** – something we will use repeatedly in this course. The idea is that the best guess for the values of x_0 and σ are those that **maximise** the probability given in 2.37. Given the exponential term, we therefore want to **minimise** the exponent. This is done by simply differentiating w.r.t. to the desired variable (i.e. x_0 and σ) and setting the result equal to zero. For x_0, we get,

$$\sum_{i=1}^{N} (x_i - x_0) = 0, \tag{2.38}$$

which is simply.

$$\text{best estimate for } x_0 = \frac{\sum x_i}{N}. \tag{2.39}$$

2.5 The Bernoulli distribution

- A **Bernoulli** event is one in which the outcomes are of the yes/no variety, such as, did the coin land heads?, did the patient survive 3 years after treatment? or did the die show a 6?, etc. Due to the widely applicable nature of this type of event, Bernoulli distributions are a common feature of statistics and data analysis.

- Let's return a classic example: what are the chances of rolling a 6 with a fair die? We are going to let the variable x take the value 1 for a 6 – i.e. 'sucess' – and 0 for a failure. The probability of rolling a 6 is going to be represented by θ and, since our die is fair, this is obviously 1/6. We can then write the probability of success p given θ as,

$$p(x|\theta) = \theta^x (1-\theta)^{(1-x)}. \tag{2.40}$$

  Let's stop and this about this. Remember that $x = 1, 0$, so for success, the first term on the RHS yields 1/6 and the second term is zero. Conversely, if $x = 0$, then the first term on the RHS is zero, and the second is the simply the probability of failure – i.e. that any other number came up, which is $(1-\theta) = 5/6$. So the Bernoulli distribution is simply another way of writing the die roll in terms of successes and failures.

- Although this form of 2.40 might seem a little contrived, it actually simplifies the mathematics when we come to consider many trials, i.e. a **series of independent events**. For example, suppose we continue to roll the die another N times. Since each roll is an independent event, we can then write the probability of getting a series of events $X = \{x_1, x_2, \ldots, x_N\}$, is given by,

$$p(\{x_1, x_2, \ldots, x_N\}|\theta) = \prod_i p(x_i|\theta) \tag{2.41}$$

$$= \prod_i \theta^{x_i} (1-\theta)^{(1-x_i)} \tag{2.42}$$

  Since x_i is 1 for a 6 and 0 for any other number of the die, we can write the number of 'successes' (i.e. the times a 6 comes up) as $\nu = \sum_i^N x_i$, then we can write the probability of obtaining **any particular sequence of successes as**,

$$p(\nu|N, \theta) = \theta^\nu (1-\theta)^{(N-\nu)} \tag{2.43}$$

2.6 The Binomial Distrbution

- The Bernouli distribution will yield the probability of throwing either,

5, **6**, 3

or

5, 1, **6**,

that is, the probability of getting any one series of events to come up with a 6. Both these two series of rolls have the same probability of occurring. But what if you don't care on the order? What you're happy with either of the two events listed above, and just care about rolling one 6 (and only one!) at any point in the series? The distribution that describes this type of probability is the **Binomial Distribution**, named due to its similarity to the Binomial Series – remember your first year maths...? :)

- Although 2.43 gives the probability of getting a particular series of ν successes in N independent trials, it doesn't account for number of different ways the series can be ordered. So if we only care about having ν successes, and don't care about the order, we need to adapt 2.43 to account for the different **combinations of successes and failures** that still yield ν successes in N trials. Perhaps a simpler way to examine this problem is to consider rolling 3 dice at the same time. What is the probability of getting ν sixes, where now $\nu = \{0, 1, 2, 3\}$?

- Let's break this problem up into the different events. First, consider $\nu = 0$,

$$p(\text{not 6, not 6, not 6}) = \left(\frac{5}{6}\right)^3 \tag{2.44}$$

and then consider $\nu = 3$,

$$p(6,\ 6,\ 6) = \left(\frac{1}{6}\right)^3. \tag{2.45}$$

These were the most straightforward as there is only 1 way in which they can occur. But now let's consider ($\nu = 1$). This can occur in 3 ways:

$$p(\text{one 6 in 3}) = p(\text{6, not 6, not 6}) + p(\text{not 6, 6, not 6}) + p(\text{not 6, not 6, 6}) \tag{2.46}$$

$$= 3\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^2. \tag{2.47}$$

Similarly for $\nu = 2$:

$$p(\text{two 6 in 3}) = p(\text{6, 6, not 6}) + p(\text{6, not 6, 6}) + p(\text{not 6, 6, 6}) \tag{2.48}$$

$$= 3\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right). \tag{2.49}$$

The coefficients that sit in front of the θ and $\theta - 1$ terms are given by the **Binomial Coefficient**,

$$\binom{N}{\nu} = \frac{N(N-1)\cdots(N-\nu+1)}{1 \times 2 \times \cdots \times \nu} \tag{2.50}$$

$$= \frac{N!}{\nu!(N-\nu)!} \tag{2.51}$$

The **Binomial Distribution** is therefore,

$$B_{N,\theta}(\nu) = \binom{N}{\nu}\theta^{\nu}(1-\theta)^{(N-\nu)}. \tag{2.52}$$

- The mean of the Binomial Distribution are given by,

$$\hat{\nu} = \sum \nu B_{N,\theta}(\nu) \tag{2.53}$$
$$= N\theta, \tag{2.54}$$

that is, if you repeat the experiment N times, the average number of successes is simply the probability of success in any one trail times the number of trials. The standard deviation is little trickier to evaluate, but is given by,

$$\sigma_\nu = \sqrt{N\theta(1-\theta)}. \tag{2.55}$$

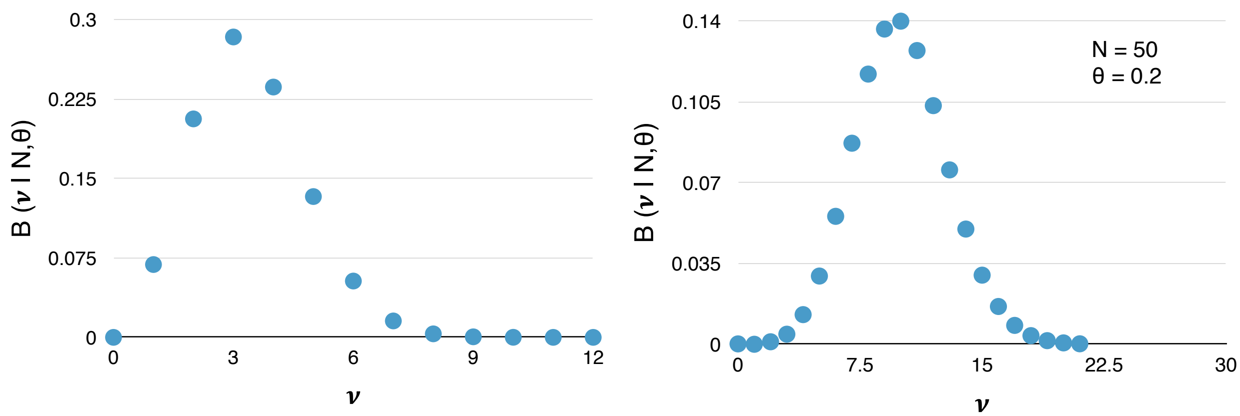

Figure 2.3: Binomial distributions for different parameters.

- Note that the Binomial Distribution distribution is only symmetric in the case where $\theta = 0.5$, as it would be in the case of the coin flip, as we can see from comparing the differences between the distributions given in Figure 2.3. However, as N increases, the distribution will start to look more symmetric, as we can see in Figure **??**.

- The Binomial Distribution will also approach the Normal distribution as $N \to \infty$. One can see this in Figure **??**, even when N is only 50. The mean and width of the Normal that the Binomial approaches are simply given by their mean and standard deviation as given above.

2.7 Beta Distributions

- The functional family that has the same form as Eq. 2.43 are called **beta distributions** and they are (usually) denoted by,

$$p(\theta|a,b) = \text{beta}(\theta|a,b) \tag{2.56}$$
$$= \frac{\theta^{(a-1)}(1-\theta)^{(b-1)}}{B(a,b)} \tag{2.57}$$

where $B(a, b)$ is the normalisation factor that ensures that the area under the curve integrates to unity, ie.,

$$B(a, b) = \int_0^1 \theta^{(a-1)}(1-\theta)^{(b-1)} d\theta. \tag{2.58}$$

Note the limits of this integral: the beta distribution is defined for θ in the range 0 to 1. Also note that the beta distribution is defined for positive a and b – zero and negative values are not defined.

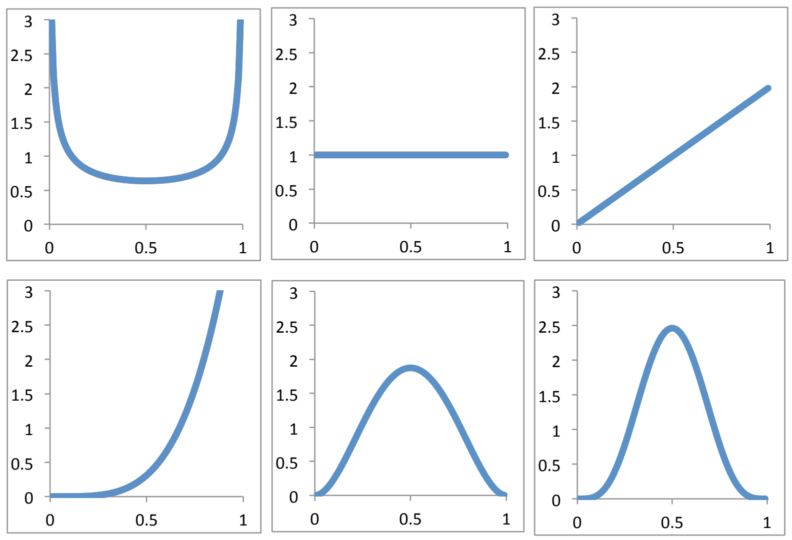

Figure 2.4: Beta distributions for various a and b

- The form of the beta distribution for different values of a and b are shown in Figure 2.4. We can see that as a gets larger, the distribution moves to high values of θ, while as b increases, the distribution moves to lower values of θ. If a and b get larger together, the beta distribution gets narrower.

- The mean and variance of the beta distribution are given by,

$$\hat{\theta}_B = \frac{a}{a+b} \tag{2.59}$$

and

$$\sigma_\sigma^2 = \frac{\hat{\theta}(1-\hat{\theta})}{a+b+1}. \tag{2.60}$$

Notice that as a becomes larger relative to b, the mean increases, and that the variance decreases as $a + b$ increases.

2.8 The Poisson Distribution

- The **Poisson Distribution** is the limiting distribution that describes the processes that are **random**, but are governed by some underlying mean rate. An example in physics include monitoring the decay of radioactive materials, or in sociology, the rate of births/deaths over winter.

- The Poisson Distribution is essentially the limit of the Binomial Distribution in the case where the number of trials is very large N is very large (think atoms in a lump of Uranium), and the probability θ is very small (the chances of a single atom decaying in a hour). In such a case, Poisson Distribution is given by,

$$P_\mu(\nu) = e^{-\mu}\,\frac{\mu^\nu}{\nu!}, \tag{2.61}$$

where μ is the mean count rate, as we are about to show, and $P_\mu(\nu)$ stands for the **the probability of ν successes (counts) in specific interval**. Derivation of the Poisson Distribution is not that difficult, but it doesn't provide any extra insight, and so we will just state it here. Basically, the motivation behind the derivation is to find a way of writing the Binomial Distribution in the limit that $N \to \infty$ and $\theta \to 0$, and expressing the limit in terms of the mean.

- We can now show that μ is simply the mean count rate. First, remember that we can write the mean number of successes in a given interval as,

$$\hat{\nu} = \sum_{\nu=0}^{\infty} \nu\, P_\mu(\nu) = \sum_{\nu=0}^{\infty} \nu e^{-\mu}\,\frac{\mu^\nu}{\nu!}. \tag{2.62}$$

Note that the first term of the sum is now zero (thanks to the leading ν) so it can be dropped. The factor $\nu/\nu!$ can be replaced by $1/(\nu-1)!$, and we can also remove a factor $\mu e^{-\mu}$, which leaves,

$$\hat{\nu} = \mu e^{-\mu} \sum_{\nu=1}^{\infty} \frac{\mu^{\nu-1}}{(\nu-1)!}. \tag{2.63}$$

The infinite sum is simply the definition of e^μ, which then cancels with the $e^{-\mu}$ out front, leaving,

$$\hat{\nu} = \mu. \tag{2.64}$$

Since, $\hat{\nu}$ is mean number of successes in a given number of trails, this proves that μ is simply the mean count rate.

- The standard deviation of the Poisson Distribution is given by,

$$\sigma_\nu = \sqrt{\mu}. \tag{2.65}$$

This is neat result: if we measure a given number of counts $\hat{\nu}$ in a given interval, then the uncertainty on our count rate is simply $\sqrt{\hat{\nu}}$

- Once again, the Normal distribution can be used as an approximation to the mean, this time in the limit that μ is large. Also note that the once again, the Poisson Distribution is **not** symmetric about the mean.

Chapter 3

A review of the addition of errors and introduction to classical hypothesis testing

3.1 Weighted Averages in Best Estimates

- Suppose we have two students, let's call them A and B, who make a measurement of the length of our snake, x. Student A finds the length to be $x = x_A \pm \sigma_A$, while student B finds that $x = x_A \pm \sigma_A$. Given that both sets of data are valid estimates of the snake's length, we'd like to combine the results from the two experiments, to get a new, and hopefully improved result x_{AB}, with an associated uncertainty σ_{AB}.
- How to proceed? It is tempting to simply average the two results, e.g. $x_{AB} = \frac{x_A + x_B}{2}$, but feels a bit fishy if the two uncertainties σ_A and σ_B are not equal. Why should they have equal weighting, if one is less accurate (higher uncertainty) than the other? The answer is to weight the values according to their uncertainties, to produce a **weighted average**.
- To work out the maths of this, we are going to assume once again that the errors in the snake length are normally distributed, and the two experiments performed by students A and B were completely independent (e.g. the snake was not stretched by A during her attempt at measurement). In that case, the probability that the students would obtain their resulting lengths for the snake is given by,

$$P_{x_0}(x_A) \propto \frac{1}{\sigma_A} e^{-(x_A - x_0)^2/2\sigma_A} \tag{3.1}$$

for student A and

$$P_{x_0}(x_B) \propto \frac{1}{\sigma_B} e^{-(x_B - x_0)^2/2\sigma_B} \tag{3.2}$$

for student B. Note that the probabilities depend on the unknown, but true value of the snake's length x_0. So the probability that **both** students found the lengths x_A and x_B is then simply,

$$P_{x_0}(x_A \cap x_B) = P_{x_0}(x_A, x_B) = P_{x_0}(x_A) \times P_{x_0}(x_B) \tag{3.3}$$

$$\propto \frac{1}{\sigma_A \sigma_B} e^{-\chi^2/2}, \tag{3.4}$$

where we have introduced the notation χ^2 ('chi-squared') as a shorthand for,

$$\chi^2 = \left(\frac{x_A - x_0}{\sigma_A}\right)^2 + \left(\frac{x_B - x_0}{\sigma_B}\right)^2. \tag{3.5}$$

Using the principle of **maximum likelihood**, we can see that $P_{x_0}(x_A, x_B)$ has a maximum when χ^2 has a minimum. So we want to know the value of x_0 that would maximise the chances of A finding x_A **and** B finding x_B. To do this, we need to differentiate χ^2 and set the derivative equal to zero,

$$2\frac{x_A - x_0}{\sigma_A} + 2\frac{x_B - x_0}{\sigma_B} = 0 \tag{3.6}$$

The solution for x_0 is then simply,

$$\text{best estimate for } x_0 = \left(\frac{x_A}{\sigma_A^2} + \frac{x_B}{\sigma_B^2}\right) \bigg/ \left(\frac{1}{\sigma_A^2} + \frac{1}{\sigma_B^2}\right) \tag{3.7}$$

If we define **weights** to have the form $w_A = {}^1/_{\sigma_A}{}^2$ and $w_B = {}^1/_{\sigma_B}{}^2$, then we can tidy this up to obtain,

$$\hat{x}_0 = \frac{w_A x_A + w_B x_B}{w_A + w_B} \tag{3.8}$$

where $\hat{x}_0$ denotes the **weighted average**.

- Using the standard error propagation formula that we covered above, we can then derive the uncertainty in $\hat{x}_0$, as,

$$\hat{\sigma}_{x_0} = \frac{1}{\sqrt{\sum w_i}} \tag{3.9}$$

where w_i denotes the individual weights of each component in the average.

- This type of weighting – also called optimal weighting – is extremely important in data analysis, and will be used later in the course, so take note! Optimal weighting allows you to take account of all data points, with each point contributing to the final result in a way than depends on how well you trust the data (i.e. the variance of the point). The problem is, that you need to know something about the error in each point (not always the case).

3.2 Error analysis

- During your first and second year labs, you will have learnt how to identify sources of error in your experiments, and how to propagate these errors through to the final result. We will briefly discuss the maths underlying the error propagation here, since it provides the background for an important property in statistics: the covariance.

- Let's first assume that we have a function f that is dependent on some measured quantity x, and yields a value y that we are interested in knowing, such that $y = f(x)$. Now the measurements of x are associated with some random error, σ_x, and so the final value of y will also have an error σ_y. How do we calculate σ_y?

- Assuming the errors in x are small, and are close to the true value $\hat{x}$, we can expand $f(x)$ around the point $\hat{x}$,

$$f(x) = f(\hat{x}) + (x - \hat{x}) \left(\frac{df}{dx}\right)_{\hat{x}} + \cdots \tag{3.10}$$

If we now identify $\hat{y} = f(\hat{x})$, then we can see that,

$$y - \hat{y} = f(x) - f(\hat{x}) \approx (x - \hat{x}) \left(\frac{df}{dx}\right)_{\hat{x}} . \tag{3.11}$$

which gives us an expression for how the value of y derived from our measured value of x, relates to the 'true' values of both y and x, which are given by $\hat{y}$ and $\hat{x}$. If we then take many measurements of x, we can use the expression above to write the standard deviation about the mean, as

$$\frac{1}{N}\sum_i^N (y_i - \hat{y})^2 = \left(\frac{df}{dx}\right)_{\hat{x}}^2 \frac{1}{N}\sum_i^N (x_i - \hat{x})^2 \tag{3.12}$$

or simply,

$$\sigma_y^2 = \left(\frac{df}{dx}\right)_{\hat{x}}^2 \sigma_x^2 \tag{3.13}$$

which is the result you are probably familiar with from your first year labs!

- In the above treatment of the Taylor expansion we ignored the second order terms and higher. An assumption here is that these terms are small (or better, exactly zero if the second derivative is 0), and if that is true, then $\hat{f}(x) = f(\hat{x})$. However, for nonlinear functions, this is not always the case, and we need to consider the higher order terms. Typically we only have to focus or attention on the second order term, which we will do so now.

- Once again, expand the function around $x = \hat{x}$:

$$f(x) = f(\hat{x}) + (x - \hat{x}) \left(\frac{df}{dx}\right)_{\hat{x}} + \frac{1}{2}(x - \hat{x})^2 \left(\frac{d^2 f}{dx^2}\right)_{\hat{x}} + \cdots . \tag{3.14}$$

Now we want to take the mean of this expression, to evaluate $\hat{f}(x)$ The mean of the first term on the RHS is simply itself, and so it doesn't change ($^1/_N \sum_N f(\hat{x}) = f(\hat{x})$). The mean of the second term is zero, since,

$$\frac{1}{N}\sum_i (x_i - \hat{x}) = \hat{x}\sum \frac{1}{N}(\frac{x_i}{\hat{x}} - 1) = \hat{x}(\frac{\hat{x}}{\hat{x}} - 1) = 0. \tag{3.15}$$

The mean of the last term contains the definition of the variance of x, such that we can write,

$$\hat{f}(x) = f(\hat{x}) + 0 + \frac{1}{2}\left(\frac{d^2 f}{dx^2}\right)_{\hat{x}} \sigma_x^2. \tag{3.16}$$

We call this last term on the RHS the **bias**. Even relatively simply functions can have a non-zero bias, for example when we translate between frequency and wavelength!

- But what if your desired quantity z is a function of more than one variable, e.g $z = f(x, y)$? Basically we proceed in the same way! First, we expand our function f around the 'true' values of $\hat{x}$ and $\hat{y}$, to get,

$$\hat{z} = \hat{f}(x,y) = f(\hat{x},\hat{y}) + \left(\frac{\partial f}{\partial x}\right)_{\hat{x}}(x - \hat{x}) + \left(\frac{\partial f}{\partial y}\right)_{\hat{y}}(y - \hat{y}) + \cdots \tag{3.17}$$

$$(z - \hat{z})^2 = (f(x,y) - f(\hat{x},\hat{y}))^2 \tag{3.18}$$

$$\approx \left(\frac{\partial f}{\partial x}\right)^2 (x - \hat{x})^2 + \left(\frac{\partial f}{\partial y}\right)^2 (y - \hat{y})^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}(x - \hat{x})(y - \hat{y}), \tag{3.19}$$

which then gives us the result that,

$$\sigma_z^2 = \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\sigma_{xy}. \tag{3.20}$$

- Ignoring the last term on the RHS of 3.20 for a moment, we see that the expression is simply the **general error propagation formula** that you learnt during your lab work. Obviously, the above analysis can be extended to any function with any number of variables $X = x_1, x_2, \ldots$, rather than just the two variables that we show here. What is important for the error propagation formula that you learn in the labs, is that the errors in the variables are **independent**, i.e. there is no correlation between the values of σ_x and σ_y.

- But what if that is not true, and σ_x and σ_y **are** correlated? That's where the last term comes in! The last term in 3.20 contains the parameter

$$\sigma_{xy} = \frac{1}{N}\sum (x - \hat{x})(y - \hat{y}), \tag{3.21}$$

which is called the **population covariance**. For truly independent variables, σ_{xy} will be zero. This is because each term in the sum can be both positive and negative, and so if the errors are independent and randomly distributed, the numerator will not accumulate as the individual measurements are added in the sum. Also, note that the denominator has an N, which ensures the sum will tend to zero. However, when the σ_x and σ_y are correlated, the numerator will grow, and thus the covariance will likely have a non-zero value.

Note that the sample covariance is given by

$$\sigma_{xy} = \frac{1}{(N-1)} \sum (x - \hat{x})(y - \hat{y}), \tag{3.22}$$

- Sometimes it can be difficult to conclude where two sources of error (or two parameters) are indeed correlated – often this is the case when the number of data points is small. So what then? If we assume the errors are independent, can we find a way to get the upper limit on the error, to ensure that we don't miss anything? The **Schwarz** inequality states that,

$$|\sigma_{xy}| \leq \sigma_x \sigma_y. \tag{3.23}$$

This allows us to rewrite 3.20 above as,

$$\sigma_z^2 \leq \left(\frac{\partial f}{\partial x}\right)^2 \sigma_x^2 + \left(\frac{\partial f}{\partial y}\right)^2 \sigma_y^2 + 2\left|\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}\right| \sigma_x \sigma_y \tag{3.24}$$

$$= \left[\ \left|\frac{\partial f}{\partial x}\right| \sigma_x + \left|\frac{\partial f}{\partial y}\right| \sigma_y\ \right]^2 \tag{3.25}$$

and so,

$$\sigma_z \leq \left|\frac{\partial f}{\partial x}\right| \sigma_x + \left|\frac{\partial f}{\partial y}\right| \sigma_y. \tag{3.26}$$

which is an upper bound to the errors in the case when the errors are **not** independent.

3.2.1 Measuring Correlations

So how do we measure the degree of correlation (the association between the observed values of two variables) in data? In almost any business, or modelling that happens, it is useful to express one quantity in terms of its relationship with others. For example, sales might increase when the marketing department spends more on TV advertisements. Often, correlation is the first step to understanding relationships between variables and subsequently building better business and statistical models.

Two variables may have a positive association, so that as the values for one variable increase, so do the values of the other variable. Alternatively, the association could be negative or neutral. Correlation quantifies this association, often as a measure between the values -1 to 1 for perfectly negatively correlated and perfectly positively correlated. The calculated correlation is referred to as the "correlation coefficient." This correlation coefficient can then be interpreted to describe the measures.

For example, suppose a Professor wants to show her students that doing well in coursework correlates with a good score in the end-of-term exam. She might then use the test data from the previous year to make a plot of homework scores against final exam scores, such as that shown in Figure 3.1. Note that the data points have no error, since the test results are essentially exact. The error here, stems from the question: are the data correlated?

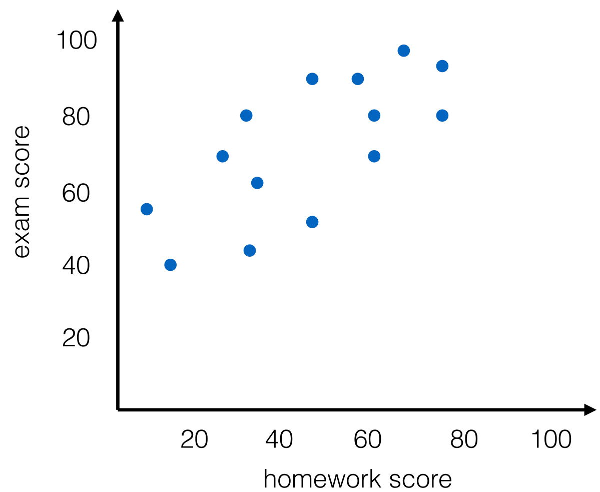

Figure 3.1: Exam scores plotted against the class homework scores

For a **linear** function, the extent to which data points $(x_1, y_1)...(x_N, y_N)$ support a linear correlation is given by the **linear correlation coefficient** sometimes called the Pearson correlation coefficient,

$$r = \frac{\sigma_{xy}}{\sigma_x \sigma_y} \tag{3.27}$$

$$= \frac{\sum(x-\hat{x})(y-\hat{y})}{\sqrt{\sum(x-\hat{x})^2 \sum(y-\hat{y})^2}}. \tag{3.28}$$

If r is close to ± 1, then the points are well correlated and the Prof's hypothesis about the homework scores is good. If it's close to zero, then the correlation is bad, and the homework scores are independent of the exam scores (see Figure3.2).

To see why $-1 < r < +1$ gives us the strength of the correlation, first note that from the Schwarz-inequality 3.23, we see that $|r| \leq 1$ or $-1 \leq r \leq 1$. Now imagine that all the points do indeed lie **exactly** on a straight line $y = A + Bx$. Since $y_i = A + Bx_i$ and $\hat{y} = A + B\hat{x}$, then $y_i - \hat{y} = B(x_i - \hat{x})$. Using this to remove the ys from 3.27, we get,

$$r = \frac{B\sum(x_i-\hat{x})^2}{\sqrt{\sum(x_i-\hat{x})^2 B^2 \sum(x-\hat{x})^2}} = \frac{B}{|B|} = \pm 1. \tag{3.29}$$

However in the case that there's no correlation with x and y, then although the numerator will fluctuate $+/-$ 've, the dominator will always be positive and drive r to zero as the number of points tend to infinity.

TABLE 7.1

Correlation Coefficient for a Direct Relationship	Correlation Coefficient for an Indirect Relationship	Relationship Strength of the Variables
0.0	0.0	None/trivial
0.1	−0.1	Weak/small
0.3	−0.3	Moderate/medium
0.5	−0.5	Strong/large
1.0	−1.0	Perfect

Figure 3.2: Table of how to interpret correlation coefficient values.

So how 'close to 1' is close enough? It turns out it is actually possible to work out the probability that r will exceed a given value r_0 after a given number of uncorrelated data points are considered, i.e. $P_N(|r| \geq r_0)$ As you imagine, the maths can be tricky, but for the Prof's case of a straight line, it turns out that with only 10 data points, $P_N(|r| \geq 0.8) = 0.005$ (i.e. 0.5%), so in her case, if she gets a value for r of around 0.8 or higher, she can be pretty confident that her hypothesis (that students who do well in coursework do well in the exam) is correct, even when she only has the data from the 10 students that took the course last year!

As an aside, correlation does not mean causation. A classic example is the correlation observed in the US between the rates of violent crime and murder and icecream, wherein the number of violent crimes spike when ice cream sales increase. But, presumably, buying ice cream doesn't turn you into a killer? This could be simply due to the fact that more ice creams are sold on sunny days ie the causing factor here is weather, not icecream.

Another classic example is the correlation observed between smoking and lung cancer in the 1950s. The correlation was observed, but to prove one led to the other is not so simple. It couold be related to those in polluted areas smoking more etc. Ultimately it took a study involving more than 40,000 doctors in the UK to show conclusively that smoking really does cause cancer.

An interesting study is described at https://mentalfloss.com/article/82108/10-times-correlation-was-not-causation by writer Maggie Koerth-Baker. She writes that a 2008 study demonstrated a correlation between the death of the Pope and when the Welsh rugby team wins the Grand Slam. This analysis appeared in the British Medical Journal's annual Christmas issue wherein they found no connection between the Pope's mortality and wins from teams from other countries. Just Wales.

A standard way of expressing the covariance is in a **variance-covariance** matrix, which we'll denote by $\mathbf{V}$. The variances appear along the diagonal and covariances appear in the off-diagonal elements, as shown below.

$$\mathbf{V} = \begin{bmatrix} \frac{\sum x_1^2}{N} & \frac{\sum x_1 x_2}{N} & \dots & \frac{\sum x_1 x_c}{N} \\ \frac{\sum x_2 x_1}{N} & \frac{\sum x_2^2}{N} & \dots & \frac{\sum x_2 x_c}{N} \\ \dots & \dots & \dots & \dots \\ \frac{\sum x_c x_1}{N} & \frac{\sum x_c x_2}{N} & \dots & \frac{\sum x_c^2}{N} \end{bmatrix} \tag{3.30}$$

where $\mathbf{V}$ is a $c \times c$ variance-covariance matrix,
N is the number of scores in each of the c data sets,
x_i is a deviation score from the ith data set ie $x - \hat{x}$,
$\sum x_i^2/N$ is the variance of elements from the ith data set.

Suppose $\mathbf{X}$ is an $n \times k$ matrix holding raw data. We can get the covariance matrix via the following matrix manipulations.

- First we transform the data matrix $\mathbf{X}$ to a 'deviation' matrix $\mathbf{x}$ such that

$$\mathbf{x} = \mathbf{X} - \mathbf{1}\mathbf{1}^{\mathsf{T}}\mathbf{X}(1/n) \tag{3.31}$$

  where n is the number of rows in the data matrix $\mathbf{X}$,
  $\mathbf{1}$ is an $n \times 1$ column vector of ones.
  T is the transpose of matrix $\mathbf{1}$. To transpose a matrix $\mathbf{A}$, row 1 of matrix $\mathbf{A}$ becomes column 1 of A^T; row 2 of $\mathbf{A}$ becomes column 2 ofA^T; and row 3 of $\mathbf{A}$ becomes column 3 of A^T. This is also sometimes denoted as A'.

- Then we compute $\mathbf{x}^{\mathsf{T}}\mathbf{x}$, the $k \times k$ deviation sums of squares and cross products matrix for $\mathbf{x}$.

- Then we divide each term in the deviation sums of squares and cross product matrix by n to create the variance-covariance matrix:

$$\mathbf{V} = \mathbf{x}^{\mathsf{T}}\mathbf{x}(1/n) \tag{3.32}$$

Note that the correlation between two variables that each have a Gaussian distribution can be calculated using standard methods such as the Pearson's correlation but this procedure cannot be used for data that does not have a Gaussian distribution. Instead, we will see something at the end of the Block which can allow us to look for correlations without assuming a Gaussian distribution.

3.3 Hypothesis Testing: the Classical Approach

Testing a hypothesis is one of the foundations of data analysis. Examples include, does this drug make people better? Is the dice fair? Are an observed population of low-mass galaxies consistent with the predictions from ΛCDM? Did CERN really detect the Higg's Boson? We'll start this section by outlining the formal ideas behind hypothesis testing, and then look at some classic examples.

3.3.1 Ideas behind 'classical' hypothesis testing

- The most common form of hypothesis test involves trying to find the unknown parameter θ that is part of a model $f(\theta)$. Now you might have a best guess for the unknown parameter, and an associated uncertainty, so really we're not always testing if θ is an exact value, but more generally whether $\theta \in \Theta$, that is θ is part of some set of possible values Θ. From our best guess of θ, what we're trying to determine is whether $\theta \in \Theta_0$ or $\theta \in \Theta_1$, and where,

$$\Theta_0 \cup \Theta_1 = \Theta \quad \text{and} \quad \Theta_0 \cap \Theta_1 = 0$$

- We then make a set of new observations of some outcome of the model $X = \{x_1, x_2, x_3, \ldots\}$, and we want to test whether they support the idea that, say, $\theta \in \Theta_1$. We also know the probability of the model predicting the data, which is given by $p(X, \theta)$.

- Classical hypothesis testing is then based around two concepts,

$$H_0 : \qquad \theta \in \Theta_0 \quad \text{the } \textbf{null} \text{ hypothesis} \tag{3.33}$$

$$H_1 : \qquad \theta \in \Theta_1 \quad \text{the } \textbf{alternative} \text{ hypothesis} \tag{3.34}$$

  The **null** hypothesis is what we are going to assume is true. We are normally trying to show that it is not!

- It is possible to make two types of error in classical hypothesis testing, and they have well defined names:

  1. A **Type I error** is when you (for some reason) reject H_0 when it is true.
  2. A **Type II error** is when you decide not to reject H_0 when it is false.

  Note – and this is extremely important – you cannot prove that something is correct in classical hypothesis testing, only prove that it is wrong. This is why the errors focus on H_0 – at best you can (correctly) accept that H_0 is correct, and thus our hypothesis that $\theta \in \Theta_1$ is wrong.

- Tests are decided based on a **rejection region**, R, such that if we observe X to lie within this region, we would reject H_0. Statisticians typically select a probability level of 5% to define the boundary of acceptability. If we find a probability below this boundary, then we can reject the null hypothesis and say that the result of the experiment is *significant*. If the probability is above the boundary then we cannot reject the null hypothesis.

- A classical statistician would then argue that decisions between tests should be based on the probability of Type I errors, i.e.

$$p(R|\theta) \; ; \; \theta \in \Theta_0 \tag{3.35}$$

and of Type II errors,

$$p(R^c|\theta) = 1 - p(R|\theta) \; ; \; \theta \in \Theta_1 \tag{3.36}$$

Generally we are trying to avoid a Type II error (i.e. falsely throwing away your idea that $\theta \in \Theta_1$, when in fact it might still be true), so we make the probability of a Type II error as small as we can, subject to the requirement that the probability of the Type I error is always less than some fixed quantity α, which is known as the **size of the test**.

- The general framework that we have outlined here is referred to as **Null Hypothesis Significance Testing**, which we will abbreviate as **NHST**. This probability is called the **p-value** of the test.

The p-value is the probability of getting a value of the test statistic at least as extreme as the actually observed value purely by chance, if the null hypothesis is true.

To summarise, we can assign a probability to our Type I error ie what is the significance level at which we reject the null hypothesis. We often choose a value of $\alpha = 0.05$ but it can be determined by the individual or team. When a p-value is less than or equal to the significance level, you reject the null hypothesis. As an example, if a test gives the p-value, $p = 0.03$, the null hypothesis would be rejected at significance level $\alpha = 0.05$, but not at the more conservative significance level $\alpha = 0.01$ (see Figure 3.3).

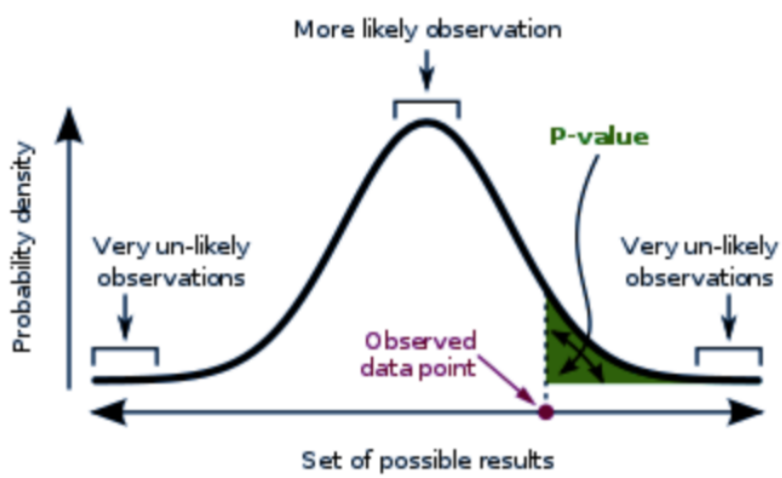

Figure 3.3: An illustration of p-value and probability distributions of data.

Chapter 4

Non-parametric statistical tests of data

A non parametric test (sometimes called a distribution free test) does not assume anything about the underlying distribution (for example, that the data comes from a normal distribution). Non-parametric tests are important particularly if the sample size is small or there are large outliers in the data.

Here we review some of the more common examples of non-parametric tests, first looking at **univariate tests** – Kolmogorov-Smirnov and Anderson Darling – and then looking at 2 **bivariate** tests – Kendall's τ and the Spearman's ρ_s rank test.

4.1 The generalised χ^2 test

- The χ^2 test is a commonly used **goodness of fit test**.
- Suppose we wish to know the range of gun. We could perform an experiment where we fire n projectiles from the gun, and measure the distance x at which the projectiles first make contact with the ground. In such case, due to the many, random uncertainties, it is natural to assume that the distribution x will be normally spread around some central value x_0, with a standard deviation of σ, such that the group of ranges $X = x_1, x_2, \ldots, x_n = N(x_0, \sigma)$
- This is a new gun, and we know very little about it, so we have no prior knowledge of either x_0 or σ. Our first task is to estimate these from the measurements. We do this in the standard way, with,

$$\text{best estimate for } x_0 = \hat{x} = \frac{\sum x_i}{n} \tag{4.1}$$

and

$$\text{best estimate for } \sigma = \hat{\sigma} = \sqrt{\frac{\sum(x_i - \hat{x})^2}{n-1}} \tag{4.2}$$

- We now want know: is our assumption that the distances are normally distributed correct? To do this, we have to compare the expected distribution to our actual distribution, to see whether our values of x are within those we expect for a limiting distribution of the form $N(x_0, \sigma)$, which we are approximating with $N(\hat{x}, \hat{\sigma})$.

- The first problem is that because x is a continuous, rather than discrete variable, we need to split the distribution into intervals – remember that PDFs are not meaningful for a given x, but rather only make sense over some range of x. So let us split our data up into, say 4 bins, with boundaries chosen at $\hat{x} - \hat{\sigma}$, $\hat{x}$ and $\hat{x} + \hat{\sigma}$, as is shown in Figure 4.1. We will refer to the number of bins as n_{bin}

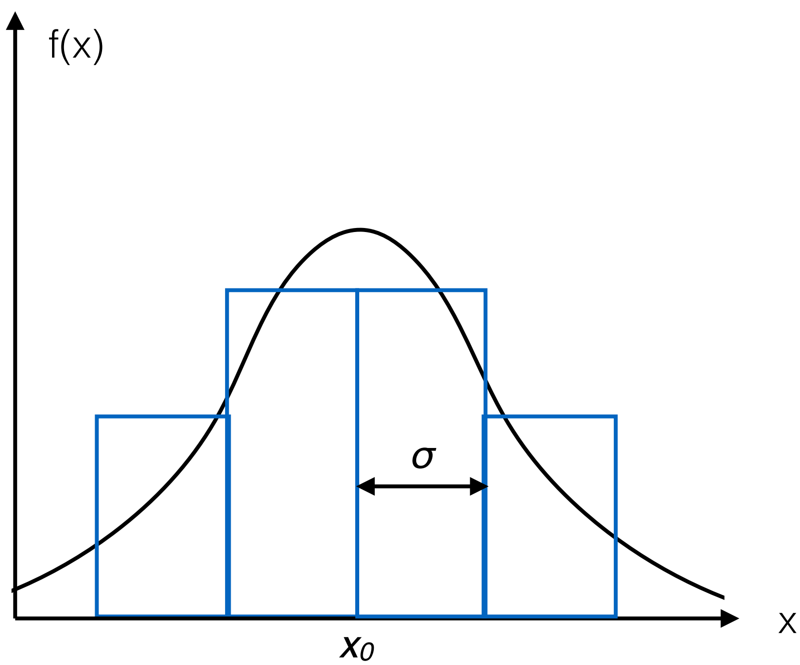

Figure 4.1:

- We can now bin our data, counting the number of measurements O_k that we observe in each bin, k. We can then compare this number to the expected number of measurements we would have obtained in the bin, E_k, if our assumptions about the limiting distribution are true. To calculate E_k for each bin, we simply work out the probability of obtaining an value in the range of x covered by the bin, and multiply this by the total number of measurements made in the experiment,

$$E_k = n \int_{x_{min}}^{x_{max}} N(\hat{x}, \hat{\sigma}) dx, \tag{4.3}$$

where x_{min} and x_{max} are the lower and upper boundaries of bin k respectively. Clearly, the 'expected number' E_k is only the number that we would expect in some average sense, and we do not expect any one experiment of finite n to yield O_k to be exactly

E_k. However we do expect the deviations to be small, if our assumption about the underlying distribution is correct.

- So how do we define 'small' or 'large' in this case? Well, the expected number of measurements in each bin can be thought of in terms of counting statistics – i.e. the Poisson distribution that we encountered before. In this sense, E_k represents the mean number that we would expect from many experiments (each taking exactly n measurements). We know that the standard deviation around this mean is given by simply $\sqrt{E_k}$. So we would expect the deviations in our observed values from the expected values, i.e. $|O_k - E_k|$, to be close to the value $\sqrt{E_k}$, if our assumptions about the underlying distribution are correct. That is, we expect,
$$\frac{|O_k - E_k|}{\sqrt{E_k}} \sim 1 \tag{4.4}$$
If we sum up the deviations in all bins, we can define,
$$\chi^2 = \sum_{k=1}^{n_{bin}} \frac{(O_k - E_k)^2}{E_k} \tag{4.5}$$
Thus if,
$$\chi^2 \leq n_{bin} \tag{4.6}$$
then we can say that there is a reasonable evidence that our assumption about the limiting function was correct! On the hand, if,
$$\chi^2 >> n_{bin} \tag{4.7}$$
then we might want to reconsider.

- Technically, one should compare the value of χ^2 with the number of degrees of freedom d, rather than the number of bins! In general,
$$d = n_{data} - n_{params} \tag{4.8}$$
where n_{data} is the number of data points, and n_{params} is number of parameters that have needed to be computed from the data – also known as the **constraints**. In the example above there were actually 3 constraints: the total number of data points n, and then the mean and variance were computed from the data. We had four bins in the pervious example, so that makes $n_{data} = 4$. So we see that $d = 1$ for this example. Had we had fewer bins, the χ^2 test wouldn't have made sense!

- This suggests that a more convenient way to think about χ^2 is via the **reduced** χ^2, or χ^2 **per degree of freedom**, which we will denote as $\tilde{\chi}^2$, and define via
$$\tilde{\chi}^2 = \frac{\chi^2}{d} \tag{4.9}$$
We then see that for the case where our assumptions about the underlying distribution are correct,
$$\text{(expected value of } \tilde{\chi}^2) \approx 1 \tag{4.10}$$

- We can also think about generalising the χ^2 expression itself. In the above example, our expected difference between the observed values and expected values was given by $\sqrt{E_k}$. However, in general we can simply write

$$\chi^2 = \sum_i^N \left(\frac{O_i - E_i}{\sigma_i}\right)^2 \tag{4.11}$$

  where obviously σ_i is the standard deviation expected for each point that is being compared to the underlying distribution.

4.2 The Kolmogorov-Smirnov Test

- The Kolmogorov-Smirnov, or 'K-S' test, is one of the most commonly used tests in data analysis. Eric Feigelson, who is on a mission to prevent people from abusing this test, noted that there are 500 uses of the K-S test every year in Astronomy/Astrophysics alone. Much of its success is due to it being easy to use!

- The K-S is used to test whether two sets of data are drawn from the same underlying limiting distribution. It is based around the **cumulative** distribution of the data. We looked cumulative distributions in Chapter 2 for continuous functions, but the principle is the same for discrete data. The cumulative distribution $S_N(x)$ of a set of N data points $\{x_1, x2, \ldots, x_N\}$ is the fraction of data points with values less than a given x. The function is created by simply sorting the data and making a running, normalised, sum for each value of x. Clearly the function moves in steps of $1/N$, being constant between consecutive values of x.

- The K-S test then measures the **maximum distance** between two cumulative distributions. If one is comparing the data set to some limiting distribution $f(x)$, with corresponding cumulative function $F(x)$, then the distance D between the two cumulative distributions is simply,

$$D = \max_{-\infty<x<\infty} |S_N(x) - F(x)| \tag{4.12}$$

  or in the case where one is comparing two data sets,

$$D = \max_{-\infty<x<\infty} |S_{N_1}(x) - S_{N_2}(x)| \tag{4.13}$$

  where $S_{N_1}(x)$ and $S_{N_2}(x)$ are the two data sets. One nice feature of using the distance in this way is that the K-S invariant to expansions and contractions in x – it works just the same for x and $log(x)$.

- The K-S test is a **null hypothesis test**, where the null is that the two distributions are **the same**. Since the K-S test relies on the distance between cumulative functions, it is mathematically possible to work out the probability of that two samples, drawn from the same underlying distributions, will have a distance D between their cumulative distributions. The maths for this involves modelling the random walk of data, in analogy to Brownian motion. The function that models this is given by

$$K(\lambda) = 2\sum_{j=1}^{\infty}(-1)^{j-1}\, e^{-2j^2\lambda^2} \tag{4.14}$$

The significance level of an observed value of D (as disproof of the null) is given approximately by,

$$\text{Prob}\,(D > \text{observed}) = K\left(\left[\sqrt{N_e} + 0.12 + 0.11/\sqrt{N_e}\right] D\right) \tag{4.15}$$

where N_e is the effective number of data points: $N_e = N$ for the case where we have a single set of data points, but given by

$$N_e = \frac{N_1 N_2}{N_1 + N_2} \tag{4.16}$$

in the case of two distributions. In Press et al., they use $j = 100$ as the limit of the sum in 4.14.

- Although the K-S test is convenient, it has a number of disadvantages.
  1. The test is not **distribution free**, and so the data cannot be used to provide the parameters for the model $f(x)$, against which it is to be tested. For example, you cannot use the the K-S test to check whether the data has been drawn from a Gaussian, for which the mean and variance was derived from the data – the model parameters need to be known in advance.
  2. Although it has been used to examine multivariate distributions, this is generally not a good idea, due to the difficulties of defining cumulative functions in multivariate space.
  3. Finally, and probably most importantly, the K-S is not good at picking up differences in the tails of functions. This is because the cumulative functions, which form the basis of the test, tend to 0 and 1 by definition. This results in the K-S test being most sensitive around the median of the distributions (the 50th percentile), and progressively less sensitive as we move to the extremes.

4.3 The Anderson-Darling test

- To improve on the problems of the K-S test in the tails of the distributions, the Anderson-Darling (A-D) test attempts to weight the data point away from the mean. The A-D statistic is given by,

$$A^2_{AD,N} = N\sum_{i=1}^{N}\frac{[i/N - S_N(x_i)]^2}{S_N(x_i)(1 - S_N(x_i))} \tag{4.17}$$

which should be computed for **both** the observed data set and the null distribution. There is unfortunately no analytic equivalent of 4.14 for the A-D test, and one instead needs to resort to numerical computation: one simply creates a suite of random draws of N points from the null distribution to build up a pdf of $A^2_{AD,N}$. The observed value of $A^2_{AD,N}$ from the data set can then be compared to the probability of drawing the same value from the null, to decide wether to retain or reject the null hypothesis that the distributions are the same.

- It has been shown (Stephens 1974) that the A-D test is significantly more powerful than the K-S test. For standard limiting distributions, such as Gaussians, etc, there also exist online tables of the critical values.
- Note that the A-D test is **distribution free** and so the properties of the null distribution can be calculated from the observed data set.

4.4 Spearman's ρ_s Rank Test

- Very often, we have a set of data N pairs of observations, e.g $(x_1, y_1), (x_2, y_2), \ldots (x_N, y_N)$, and wish to determine whether the points are correlated. We mentioned earlier that we can use Pearson's Linear Correlation Coefficient to determine whether we have linear trends in the data. However, what if the correlation is possibly non-linear? Or what if the data spans many orders of magnitude? The Spearman Rank Test gets around this by looking at the **rank** of the data: where in the sorted values X does measurement x_i appear, and how does that relate to where its partner y_i appears in the sorted values of Y?
- The rank test ρ_s is actually just the Pearson's linear correlation coefficient r, but applied to the ranks of the data:

$$\rho_s = \frac{\sum_{i=1}^{N} R(x_i)\,R(y_i) - N(N+1)^2/4}{\sqrt{\sum_{i=1}^{N} R(x_i)^2 - N(N+1)^2/4}\sqrt{\sum_{i=1}^{N} R(y_i)^2 - N(N+1)^2/4}} \tag{4.18}$$

- In the case where the data have no "ties" – i.e. values with the same x or y values, this simplifies to the following:

$$\rho_s = 1 - \frac{\sum_{i=1}^{N} [R(x_i) - R(y_i)]^2}{N\,(N^2-1)} \tag{4.19}$$

- For N larger than around 30, the null hypothesis for ρ_s is distributed as a normal, with a mean of zero and a variance of $\frac{1}{N-1}$. For smaller values of N, the null has to be explored numerically, although tables are available online.

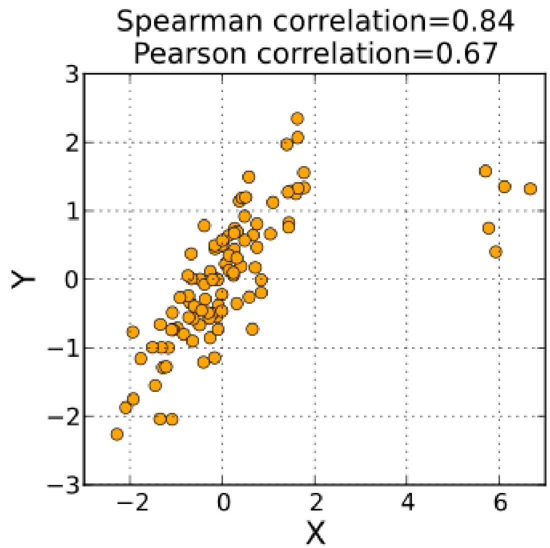

Figure 4.2:

- One nice feature of the Spearman test is that it is not so sensitive to outliers at the extremes of x and y as the Pearson correlation co-efficient. An example of this can be seen in Figure 4.2, where we see that the Spearman test is less dominated by the points at far right right of the graph. This is because the points are treated by their **rank** rather than their intrinsic value.

4.5 Kendall's τ Test

- A very similar test to Spearman's is Kendall's τ Test, which is based around the idea of **concordant** and **discordant** pairs. The concordant pairs in which both the x-values and y-values are going the same direction (either rising or falling). Ie this does not take into account the difference between ranks — only directional agreement. Therefore, this coefficient is more appropriate for discrete data. More formally:

$$\frac{y_i - y_j}{x_i - x_j} > 0. \tag{4.20}$$

  A discordant pair is one for which the opposite is true:

$$\frac{y_i - y_j}{x_i - x_j} < 0. \tag{4.21}$$

- The test statistic is then defines as,

$$\tau_K = \frac{N_c - N_d}{N_c + N_d} \tag{4.22}$$

Ties in y are assumed to account equally as a 1/2 to both N_c and N_d. Ties in x pairs are ignored. If N is large, the distribution of τ_K under the null hypothesis is then given by a normal distribution, with mean and variance:

$$E[\tau_K] = 0 \tag{4.23}$$

$$Var[\tau_K] = \frac{\sqrt{2(2N+5)}}{3\sqrt{N(N-1)}} \tag{4.24}$$